AKDENIZ UNIVERSITY
CSE 328 – Internet of Things
Term Project Proposal

Student Information

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1) Project Title: ESP32-Based Motor Fault Detection System with On-Device TinyML via Edge Impulse

2a) Executive Summary

This project proposes an ESP32-based motor fault detection system that performs real-time condition monitoring using on-device TinyML. The system uses an ADXL345 accelerometer to capture vibration signatures and an MLX90614 infrared temperature sensor to detect abnormal thermal behavior. A model trained in Edge Impulse Studio classifies motor operation into NORMAL, WARNING, and FAULT states directly on the ESP32 without requiring continuous cloud-side inference. In addition to local intelligence, the system includes an OLED screen for live status visualization and a 5V relay module to cut motor power automatically when a fault is detected. An RGB LED provides a quick visual indication of the current operating state. The cloud dashboard is used for remote monitoring of sensor values, inference outputs, and system status, while also allowing motor enable/disable and manual fault reset commands. This design aims to provide an efficient, low-cost, and safer predictive maintenance solution for small-scale industrial or laboratory motor systems.

2b) Project Description

Motor faults in industrial and hobby-scale machines rarely appear suddenly; instead, early degradation mechanisms such as bearing wear, rotor imbalance, and shaft misalignment usually emerge first as subtle changes in the machine's vibration signature. Conventional threshold-based monitoring systems are often limited to simple amplitude checks and may miss these more complex temporal and spectral patterns, especially when the abnormality is still developing and the motor has not yet reached a clearly unsafe operating region.

This project proposes an ESP32-based motor fault detection system that performs real-time classification directly on the device by running a TinyML model trained with Edge Impulse. Rather than transmitting raw vibration streams to the cloud for analysis, the ESP32 executes inference locally and classifies the motor state as NORMAL, WARNING, or FAULT, which reduces latency, lowers bandwidth requirements, and enables immediate protective action even when network connectivity is unstable or unavailable.

The primary sensing element is an ADXL345 three-axis accelerometer connected over SPI, which captures the vibration waveform of the motor and feeds a one-second sample window into the deployed inference pipeline. In Edge Impulse Studio, this vibration data is processed with spectral analysis features and used to train a compact Keras neural network, while a K-Means anomaly detection block provides an additional measure of deviation from nominal behavior; the final model is then exported as a C++ library and integrated into the ESP32 firmware for on-device execution.

To complement vibration-based diagnosis, an MLX90614 infrared thermometer connected via I2C monitors motor surface temperature as an independent overheat threshold channel. At runtime, the ESP32 combines the TinyML output and the thermal check inside a state machine: in FAULT condition, a 5V relay immediately cuts motor power, an RGB LED presents the operating state visually, and the OLED displays the inference label, confidence score, anomaly score, and temperature in live form.

For operational safety, the FAULT condition is latched into a LOCKED state so that the motor cannot restart accidentally after a severe event. Recovery requires an explicit manual reset command from the remote dashboard, making the design suitable for both instructional demonstration and basic predictive-maintenance scenarios.

Edge Impulse Studio is used as the project's cloud-side platform for dataset collection, model training, deployment artifacts, and performance metrics, while the connected dashboard provides bidirectional supervision for monitoring sensor outputs, reviewing model behavior, enabling or disabling the motor, and issuing a controlled fault reset when needed.

3) Hardware and Software Requirements Checklist

☒ ESP32 will be used

☒ 2 sensors will be used (two different sensor types)

Sensor 1 model/type and its task: ADXL345 (3-axis accelerometer, SPI) - captures motor vibration data as primary input to the on-device TinyML inference model

Sensor 2 model/type and its task: MLX90614 (non-contact IR thermometer, I2C) - monitors motor surface temperature as an independent overheat threshold check

☒ OLED display will be used.

☒ + 1 additional actuator will be used

Actuator 1 model/type and its task: 5V relay module - interrupts motor power on FAULT; re-enabled only via dashboard reset command

☒ Data will be sent to a cloud platform (Edge Impulse Studio)

☒ Dashboard will monitor both sensors

☒ Dashboard will control the actuator (bidirectional): Motor enable/disable and fault reset via Edge Impulse dashboard

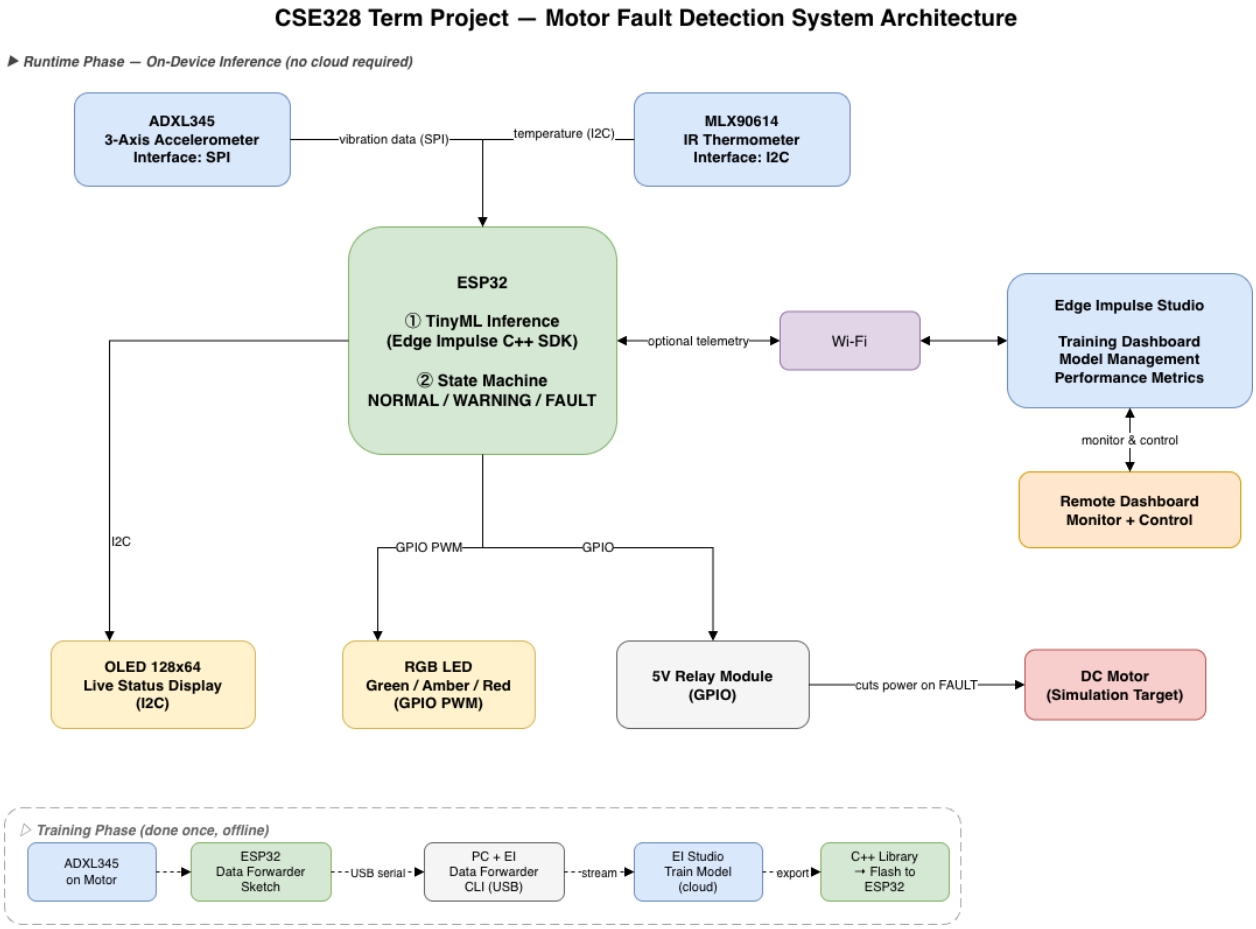
(Optional / extra feature)

☒ Dashboard will process data: per-class confidence scores, K-Means anomaly score, inference latency and RAM/flash usage displayed in Edge Impulse Studio

☒ Other extra feature: RGB LED state indicator (green = NORMAL, amber = WARNING, red blink = FAULT); LOCKED state prevents motor restart without explicit dashboard reset

4) System Architecture (diagram + short explanation)

The ESP32 operates across two phases. In the training phase (offline, done once), the ADXL345 streams vibration data via the Edge Impulse Data Forwarder to Edge Impulse Studio, where a spectral analysis block, a Keras neural network, and a K-Means anomaly detector are trained and exported as an Arduino C++ library. In the runtime phase, the library runs entirely on-device: the ADXL345 fills a one-second sample buffer, the model classifies the motor state, and the MLX90614 provides a parallel temperature check. Both outputs drive the relay and RGB LED through a state machine, while the OLED displays live results. Wi-Fi connects the ESP32 to Edge Impulse Studio for optional telemetry and remote motor control.



5) Planned Dashboard User Interface

The OLED display shows four lines of live data. Line 1: inference result and confidence score. Line 2: K-Means anomaly score. Line 3: motor surface temperature with a bar indicator. Line 4: relay state and deployed model version. In FAULT state the screen flashes a full-screen "!! MOTOR STOPPED !!" message. The Edge Impulse Studio dashboard provides remote monitoring (sensor charts, model performance) and bidirectional control (motor enable/disable, fault reset).